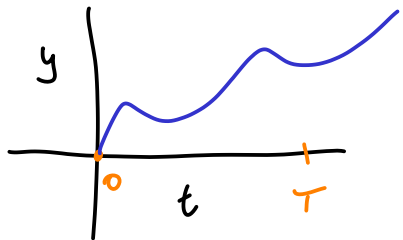


Lecture 6: System Identification

Mathias Hudoba de Badyn

October 16 2025

Recap of Feedforward Control

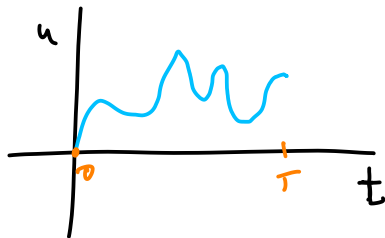


$$\dot{x} = Ax + Bu$$

$$y = Cx := y_d$$



inverse system



$$\dot{\eta} = A_{inv} \eta + B_{inv} y_d$$

$$u = C_{inv} \eta + D_{inv} y_d$$



Step 5: Complete the Inverse System

$$\begin{aligned}\dot{\eta}_{ref} &= \underline{A_{inv}} \eta_{ref} + B_{inv} \mathcal{Y}_d \\ u_{inv} &= \underline{C_{inv}} \eta_{ref} + D_{inv} \mathcal{Y}_d\end{aligned}$$

$\rightarrow A_{inv}$: evals are independent of T_η

$\rightarrow$ evals are the zeros of the original system

$$A_{inv} = \underbrace{T_\eta}_{\text{circled}} [A - (BB_y^{-1}A_y)] T_r^{-1}$$

$$B_{inv} = \begin{bmatrix} T_\eta [A - (BB_y^{-1}A_y)] T_l^{-1} & \vdots & T_\eta BB_y^{-1} \end{bmatrix}$$

$$C_{inv} = -B_y^{-1}A_y T_r^{-1}$$

$$D_{inv} = \begin{bmatrix} -B_y^{-1}A_y T_l^{-1} & \vdots & B_y^{-1} \end{bmatrix}$$

$$\mathcal{Y}_d(t) = \begin{bmatrix} \xi_d(t) \\ y_d^{(r)}(t) \end{bmatrix}$$

$\Rightarrow A_{inv}$ is unstable for non-minphase systems

Step 5.5: Solve the Inverse System

$$\left. \begin{aligned} \dot{y} &= A_{inv} \eta + B_{inv} y_d \\ u &= C_{inv} \eta + D_{inv} y_d \end{aligned} \right\} \text{ IVP} \Rightarrow \text{need } \eta(t_0)$$

assume $\dot{x}=0 \Rightarrow \dot{y}=0 = A_{inv} \eta(t_0) + B_{inv} y_d(t_0)$
invertible $\Rightarrow \eta(t_0) = -A_{inv}^{-1} B_{inv} y_d(t_0)$

Step 6: For non-minimum phase systems

NMPS $\Rightarrow$ RHP evals of inverse system

① define T_{split} that splits η into its stable/unstable modes

$$T_{\text{split}} \begin{bmatrix} \eta_s \\ \eta_u \end{bmatrix} = \eta \longrightarrow \text{MATLAB}$$

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \eta_s \\ \eta_u \end{bmatrix} &= T_{\text{split}}^{-1} A_{\text{inv}} T_{\text{split}} \begin{bmatrix} \eta_s \\ \eta_u \end{bmatrix} + \underbrace{T_{\text{split}}^{-1} B_{\text{inv}}}_{\text{}} y_d \\ &= \begin{bmatrix} A_s & 0 \\ 0 & A_u \end{bmatrix} \begin{bmatrix} \eta_s \\ \eta_u \end{bmatrix} + \begin{bmatrix} B_s \\ B_u \end{bmatrix} y_d \end{aligned}$$

Step 6: For non-minimum phase systems

Stable

$$\dot{y}_s = A_s y_s + B_s y_d$$

↳ solve this

$$y_{s,ref}(t) = \int_{-\infty}^t e^{A_s(t-\tau)} B_s y_d(\tau) d\tau$$

Sort from IVP

unstable

$$e^t \rightarrow e^{-t} \\ (t \rightarrow -t)$$

$$\dot{y}_u = A_u y_u + B_u y_d$$

↳ stable if you solve backwards in time

$$\dot{y}_u = -A_u y_u - B_u y_d$$

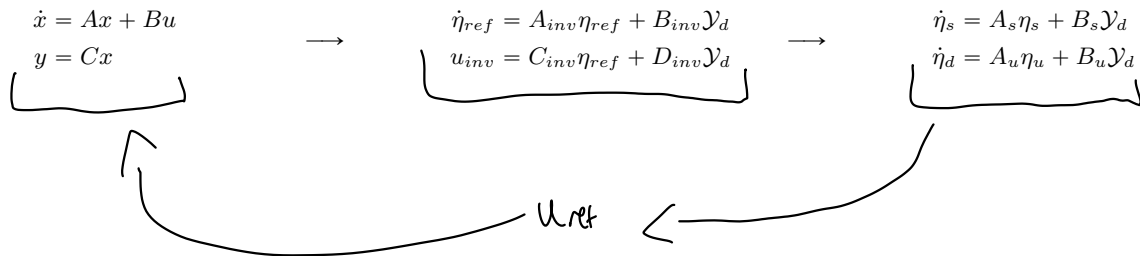
$$y_{u,ref}(t) = \int_t^{\infty} e^{-A_u(\tau-t)} B_u y_d(\tau) d\tau$$

$$w_{inv}(t) = C_{inv} y_{ref} + D_{inv} y_d \leftarrow y_{ref} = T_{split} \begin{bmatrix} y_{s,ref} \\ y_{u,ref} \end{bmatrix}$$

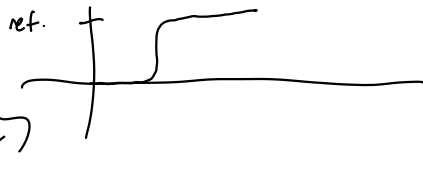
Step 6: For non-minimum phase systems

Matlab Example

Non-Minimum Phase System



Focus: Internal Dynamics



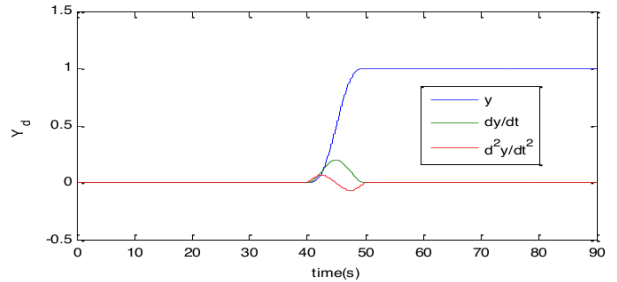
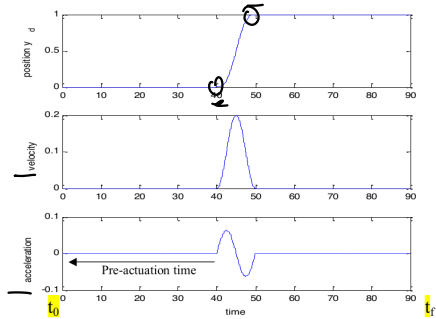
$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$

$$\dot{\eta}_d = A_u \eta_u + B_u \mathcal{Y}_d$$

$$A_s = \underline{-0.95}$$

$$\underline{A_u = 1.05}$$

Desired Trajectory



Solve Stable Subsystem

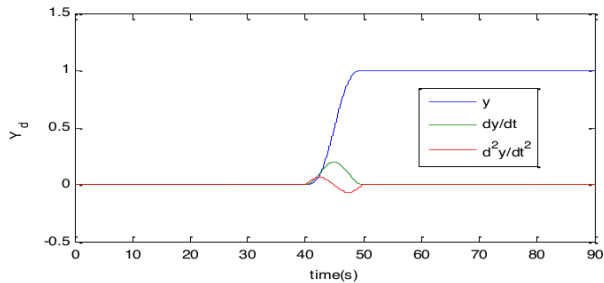
$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$

$$Y_s = C_s \eta_s + D_s \mathcal{Y}_d$$

$$C_s = [1], \quad D_s = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{Y}_d = \underbrace{\begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T}$$

comes from relative
degree of original system



Solve Stable Subsystem

$$\dot{\eta}_s = A_s \eta_s + B_s \mathcal{Y}_d$$

$$Y_s = C_s \eta_s + D_s \mathcal{Y}_d$$

$$C_s = [1], \quad D_s = [0 \quad 0 \quad 0]$$

$$\mathcal{Y}_d = [y \quad \dot{y} \quad \ddot{y}]^T$$

Initial condition (from equilibrium:)

$$0 = A_s \eta_s + B_s \mathcal{Y}_d$$

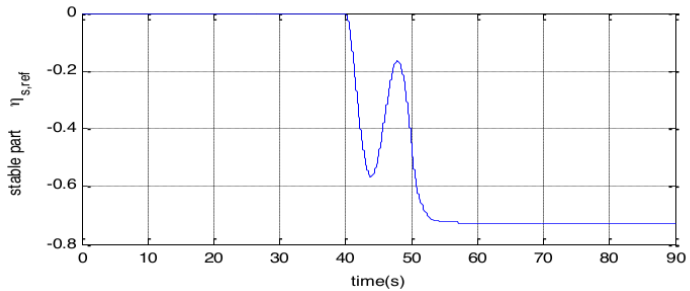
$$\eta_s(t_0) = -A_s^{-1} B_s \mathcal{Y}_d(t_0)$$

$$\text{sys_s} = \text{ss}(A_s, B_s, C_s, D_s)$$

$$[\mathcal{Y}_d, \underline{t}, \eta_{s,ref}] = \text{lsim}(\text{sys_s}, \mathcal{Y}_d, \underline{t}, \eta_s(t_0))$$

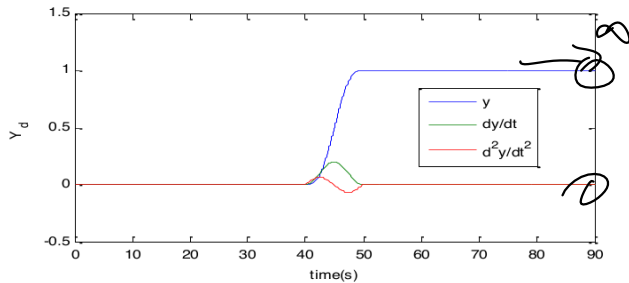
same length

Solve Stable Subsystem



Solve Unstable Subsystem

$$\begin{cases} \dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d \\ Y_u = C_u \eta_u + D_u \mathcal{Y}_d \end{cases}$$
$$C_u = [1], \quad D_u = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$
$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$



Solve Unstable Subsystem

$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$

$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

$$C_u = [1], \quad D_u = [0 \quad 0 \quad 0]$$

$$\mathcal{Y}_d = [y \quad \dot{y} \quad \ddot{y}]^T$$

Solve it backwards in time:

$$\dot{\eta}_u = -A_u \eta_u + -B_u \mathcal{Y}_d$$

$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

Solve Unstable Subsystem

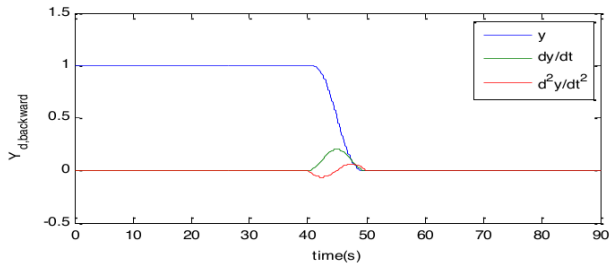
$$\dot{\eta}_u = A_u \eta_u + B_u \mathcal{Y}_d$$

$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

$$C_u = [1], \quad D_u = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\mathcal{Y}_d = \begin{bmatrix} y & \dot{y} & \ddot{y} \end{bmatrix}^T$$

Solve it backwards in time:



Solve Unstable Subsystem

$$\dot{\eta}_u = -A_u \eta_u + -B_u \mathcal{Y}_d$$
$$Y_u = C_u \eta_u + D_u \mathcal{Y}_d$$

Initial condition (from equilibrium:)

$$0 = A_u \eta_u + B_u \mathcal{Y}_d$$

$$\underline{\eta_u(t_f)} = -A_u^{-1} B_u \mathcal{Y}_d(t_f)$$

$$\text{sys_u_bw} = \text{ss}(\underline{-A_u}, \underline{-B_u}, \underline{C_u}, \underline{D_u})$$

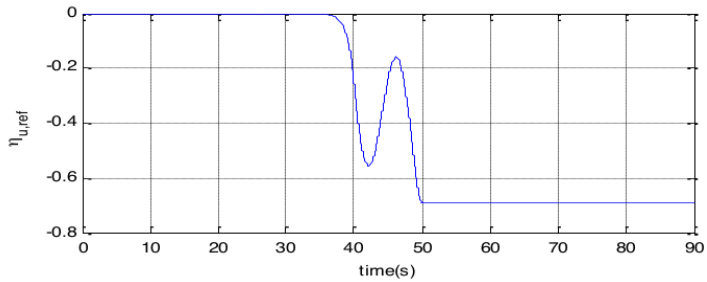
$$\mathcal{Y}_{d,bw} = \underline{\text{fliplr}(\mathcal{Y}_d)}$$

$$[\mathcal{Y}_{ub}, t, \eta_{ub}] = \underline{\text{lsim}(\text{sys_u_bw}, \mathcal{Y}_{d,bw}, \eta_u(t_f))}$$

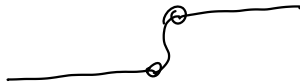
$$\eta_{u,ref} = \underline{\text{fliplr}(\eta_u^T)}$$

flipud

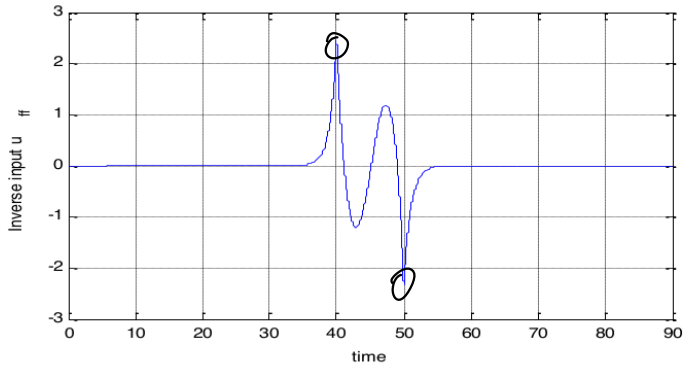
Solve Stable Subsystem



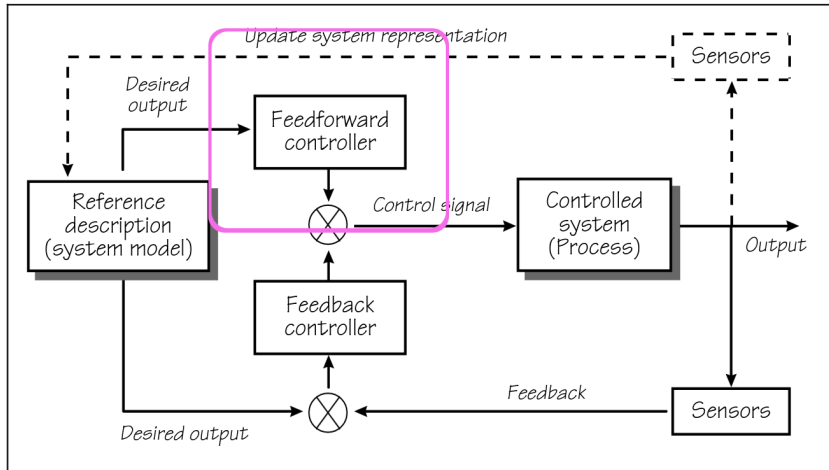
Find Reference Input



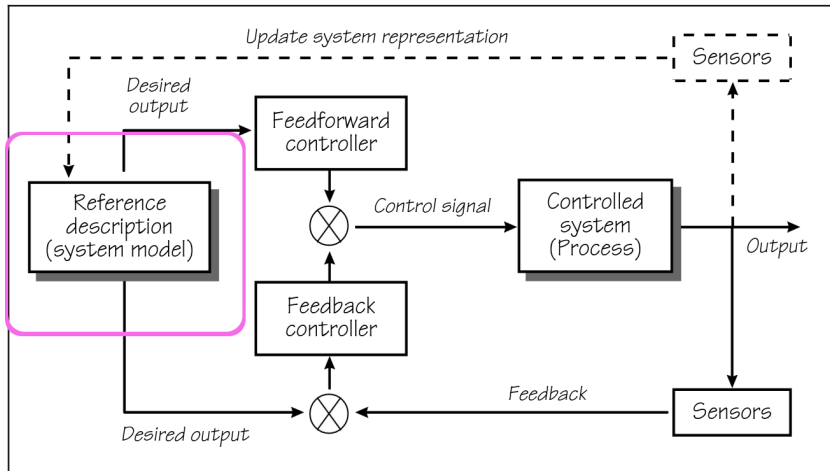
$$\eta_{ref} = T_{split} \begin{bmatrix} \eta_{s,ref} \\ \eta_{u,ref} \end{bmatrix}$$
$$u_{inv}(t) = C_{inv} \eta_{ref} + D_{inv} \mathcal{Y}_d$$



Today



Next Time



System Identification

Lecture 6: Introduction to System Identification

Mathias Hudoba de Badyn

Today

Reading assignment:

- ▶ **Ljung:** Ch6,7,10

Identification objectives

- ▶ Simple approximation to the system.
- ▶ Model for prediction purposes.
- ▶ Clearer understanding of system behaviour.
- ▶ Model for use in control design.

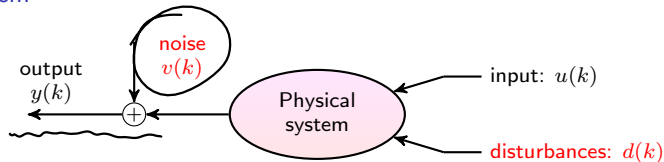
Fundamental issues: identifying dynamical systems

- ▶ Purpose of the model.
- ▶ Open- or closed-loop experiments.
- ▶ Linearity of the system. Linear model assumptions.
- ▶ Choice of sample period, T .
- ▶ Finite data records.
- ▶ Constraints on the input excitation.
- ▶ Noise on the measurements.
- ▶ Experiment time constraints.
- ▶ Data efficiency (how big is the data?)

does a ctrl already exist?

Identification framework: open-loop configuration

Experimental system

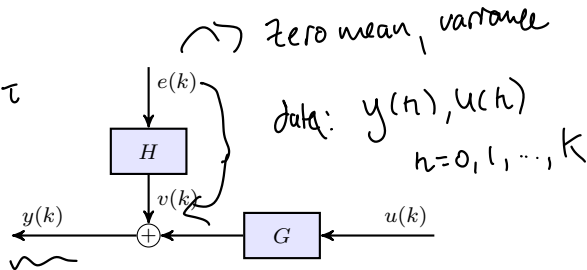


Identified model

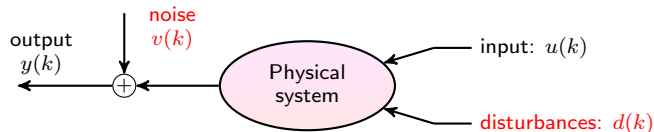
$$y(t) = \int_0^t g(\tau) u(t-\tau) d\tau$$

$$y(k) = \sum_{i=0}^n g(i) u(k-i)$$

$$Y(s) = G(s) U(s)$$



Fundamental concepts



An Identification Procedure: from data to a model

1. Choose an input of length K : $u(k)$, $k = 0, \dots, K - 1$.
2. Apply $u(k)$ to the plant, G .
3. Measure $y(k)$, $k = 0, \dots, K - 1$.
4. Use $\{u(k), y(k)\}$, $k = 0, \dots, K - 1$ to get a model, $\hat{G}$, for the plant G .

Basic idea: $y = Gu$, so $\hat{G} \approx y/u$.

Models

First principles (from the physics)

Derived from physical principles.

$$\underbrace{f = m\ddot{x}} \implies \underbrace{G(s) = \frac{1/m}{s^2}}, \text{ where } \underbrace{y = x}, \text{ and } \underbrace{u = f.}$$

Models

First principles (from the physics)

Derived from physical principles.

$$f = m\ddot{x} \implies G(s) = \frac{1/m}{s^2}, \text{ where } y = x, \text{ and } u = f.$$

“Black-box” models

Response estimated from experimental data (measurements of $\underbrace{f(t)}$ and $\underbrace{x(t)}$).

Models

First principles (from the physics)

Derived from physical principles.

$$f = m\ddot{x} \implies G(s) = \frac{1/m}{s^2}, \text{ where } y = x, \text{ and } u = f.$$

“Black-box” models

Response estimated from experimental data (measurements of $f(t)$ and $x(t)$).

“Grey-box” models

A combination of first principles and experimentally derived parameters.

$$\underbrace{G(s) = \frac{\theta}{s^2}}_{\text{where } \theta \text{ is to be estimated from data.}}$$

Modeling philosophy

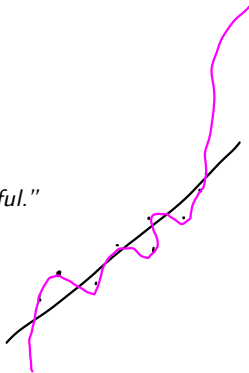
fma

Applicability: *"All models are wrong, but some are useful."*

George Box, 1978

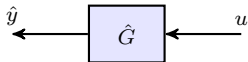
Parsimony:

- ▶ Make the model as simple as possible.
 - ▶ Occam's razor - remove extraneous assumptions.
 - ▶ Simpler models are more easily falsified.
 - ▶ Fewer parameters require less data for identification.



Model purpose

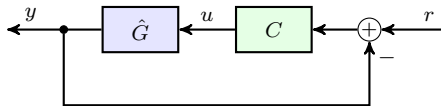
Open-loop prediction



$$\hat{y} = \hat{G}u.$$

Criterion: $\|y - \hat{y}\|$

Closed-loop design

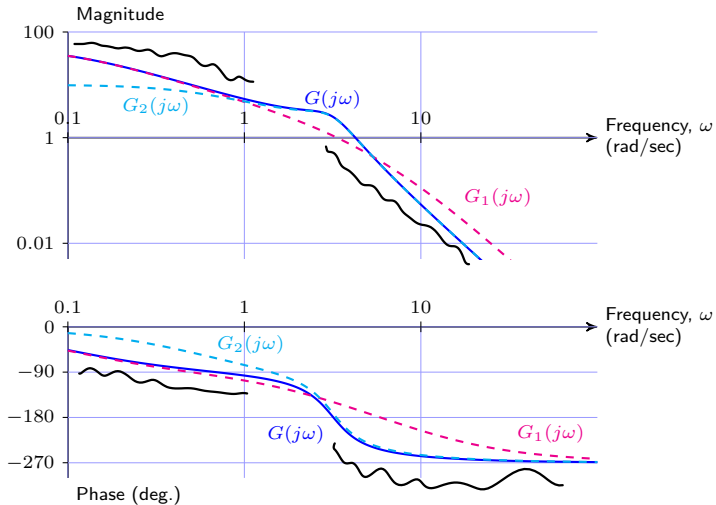


$$\frac{y}{r} = \frac{\hat{G}C}{1 + \hat{G}C}$$

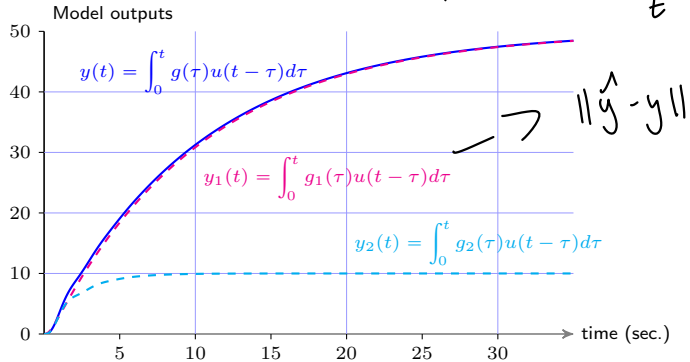
Criteria: $\frac{GC}{1 + GC}$ stable

$$\left\| \frac{GC}{1 + GC} - \frac{\hat{G}C}{1 + \hat{G}C} \right\|$$

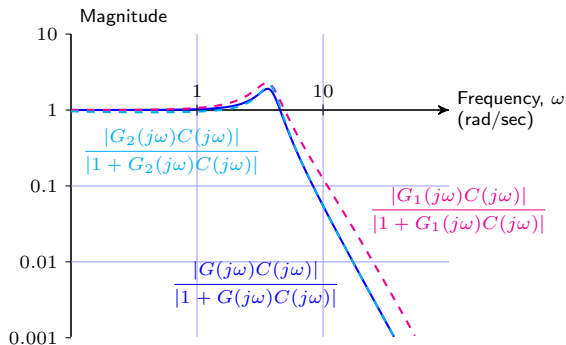
Model purpose: frequency domain



Model purpose: Step responses



Model purpose: Closed-loop responses



Controller:

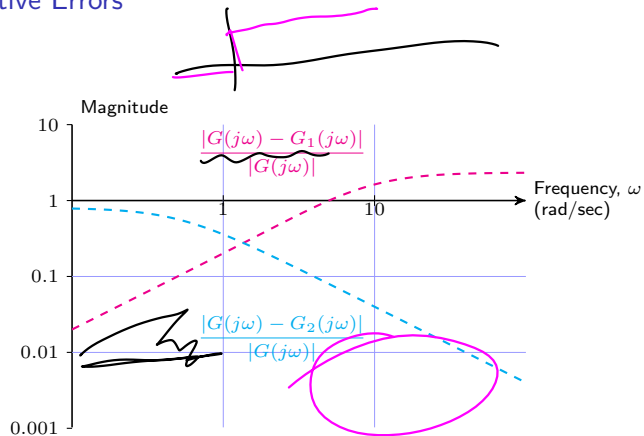
$$C(s) = \frac{0.01(10s + 1)}{(0.1s + 1)}$$

Closed-loop stability:

- G_1 stable
- G_2 unstable
- G unstable

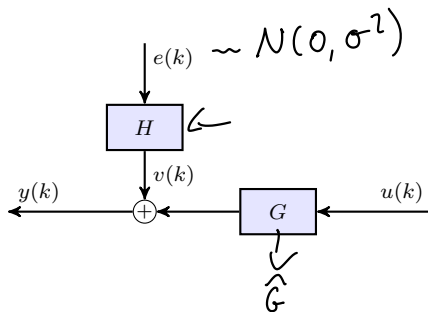
G_2 was designed to optimise the closed-loop response

Model purpose: Relative Errors



Open-loop identification

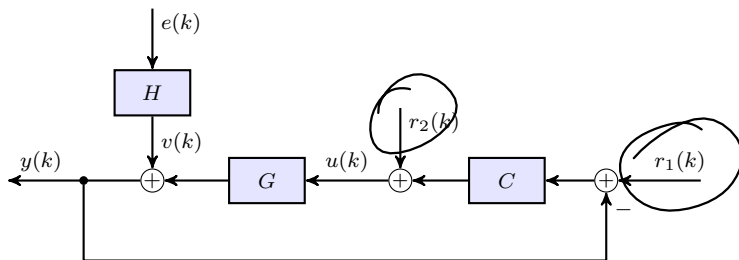
Discrete-time model:



Main assumptions

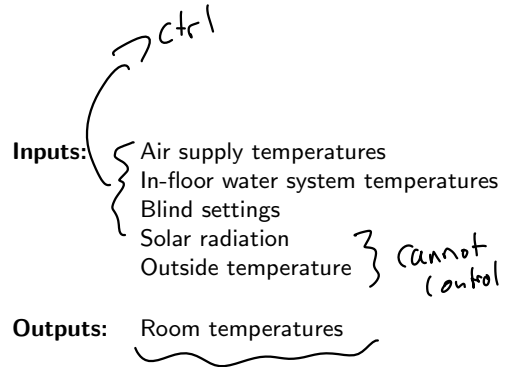
- ▶ G is linear.
- ▶ $e(k)$ is zero-mean white noise (with known variance).
- ▶ H may be known approximately.

Closed-loop identification

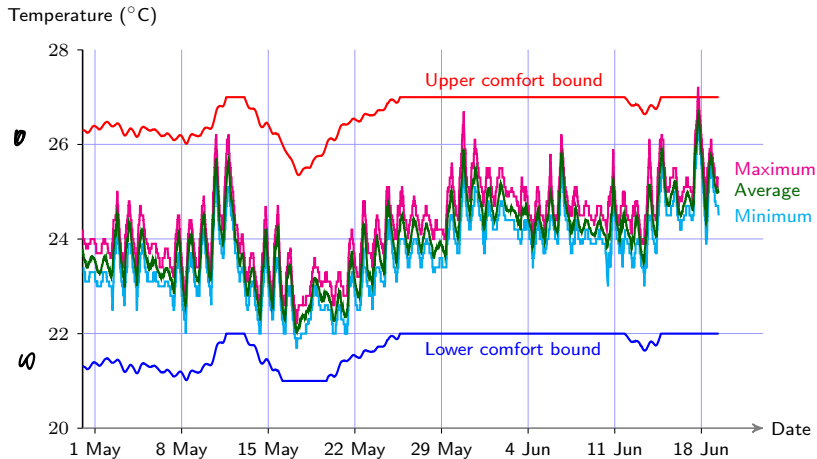


- ▶ Closed-loop operation can set the operating point of the plant.
- ▶ For many plants closed-loop operation is required.
- ▶ For unstable plants, closed-loop experiments are essential.
- ▶ A well designed controller masks variations in the plant.
- ▶ We can now measure $y(k)$, $u(k)$, $r_1(k)$ and $r_2(k)$.
- ▶ All of these signals are now correlated.

Example: energy flows in buildings



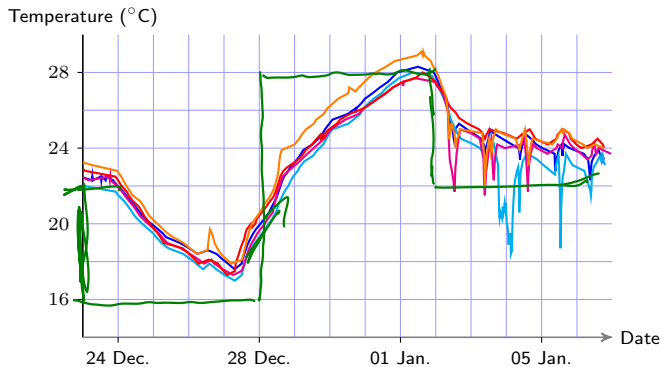
Example: energy flows in buildings – Room temperatures (controlled operation)



Step (or doublet) excitation

Step changes in heating setpoint:

Excitation plan: 16 °C for 5 days
30 °C for 5 days

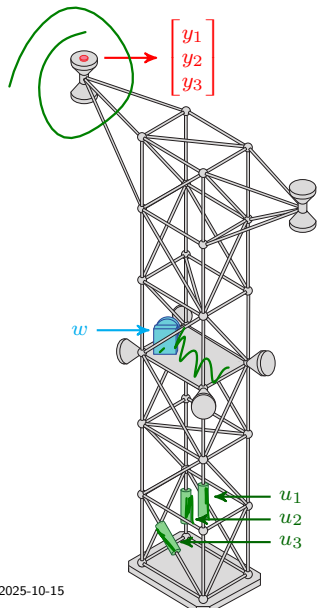


Example: energy flows in buildings

Things to consider:

- ▶ The building is occupied and (usually) controlled.
 - ▶ The dynamics can be (very) slow.
 - ▶ It takes a year to see a reasonably complete range of weather conditions.
- 

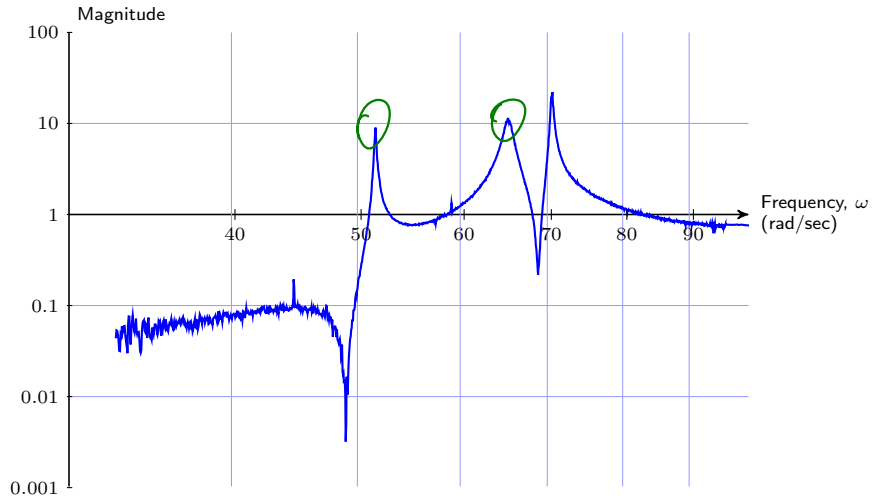
Example: vibration in flexible structures



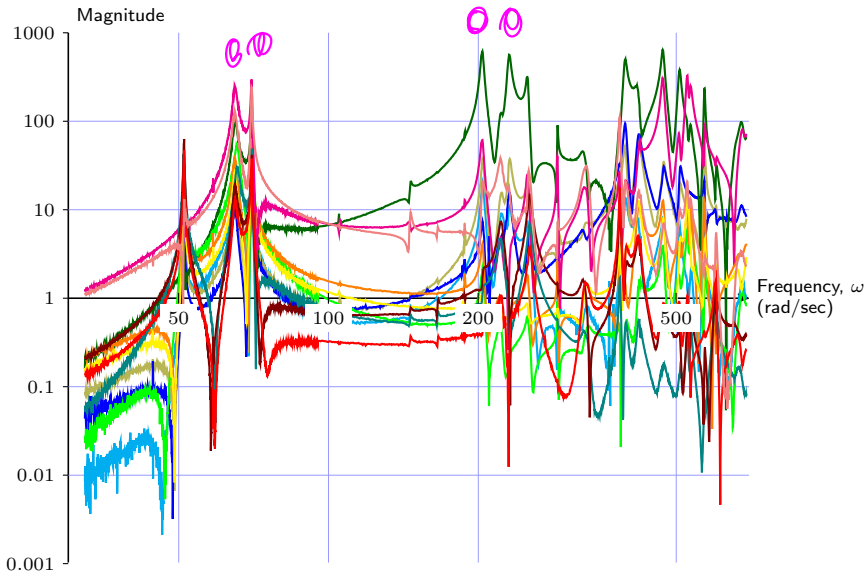
Inputs: Forces in structural members. (u_i)
Shaker disturbance (w)

Outputs: Accelerations in 3 dimensions (y_i)

Example: vibration in flexible structures



Example: vibration in flexible structures

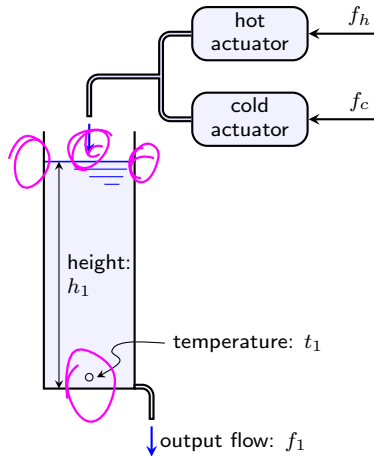
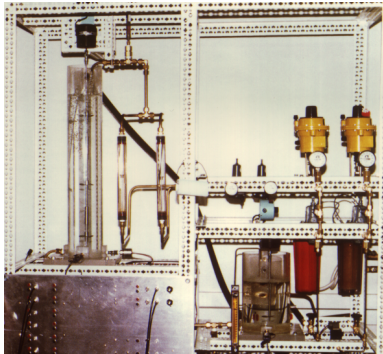


Example: vibration in flexible structures

Things to consider:

- ▶ Extremely fine frequency resolution is needed to resolve the modes.
- ▶ Long data records are required.
- ▶ There is noise and drift on the accelerometer measurements.
- ▶ Experiments are very sensitive to environmental disturbances.

Example: process control

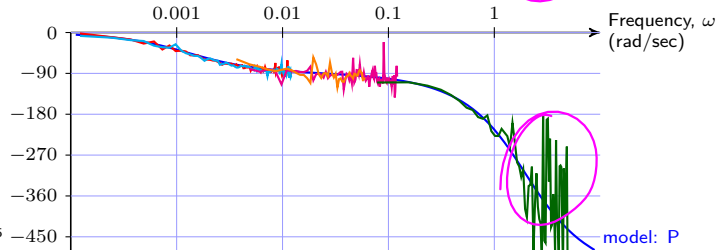
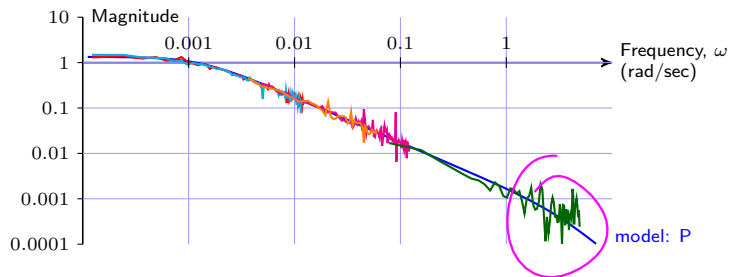


Inputs: Hot and cold water flows (f_h and f_c)

Outputs: Tank height and temperature (h_1 and t_1)

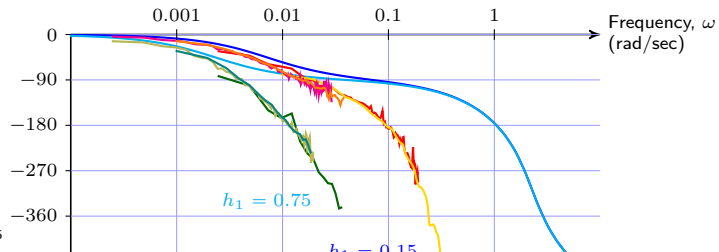
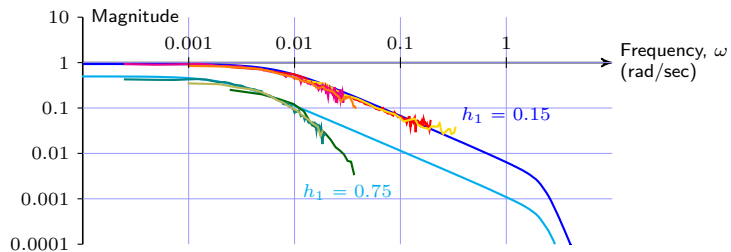
Example: process control

Height response: $f_h + f_c$ to h_1



Example: process control

Temperature response: $f_h - f_c$ to t_1



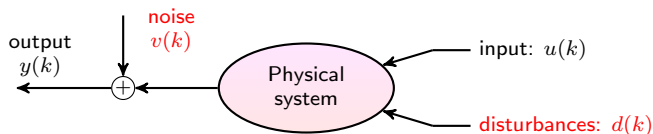
Example: process control

Things to consider:

- ▶ Temperature and pressure (height) measurements are noisy.
- ▶ A wide range of frequencies were identified (multiple experiments).
- ▶ Nonlinear mixing dynamics are significant in the temperature response.
- ▶ The temperature response is a strong function of the height.
- ▶ Closed-loop control was used for one variable (h_1 or t_1), while the other was excited in open-loop.

Identification: static open-loop case

Experimental system



Experiment:

Length K data record: $\mathcal{Z} = \{ (y(k), u(k)) \} \quad k = 0, \dots, K - 1.$

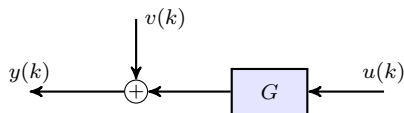
We may have multiple experiments, $\mathcal{Z}_r, \quad r = 1, \dots, R.$

Identification: static open-loop case

Experiment:

Length K data record: $\mathcal{Z} = \{(y(k), u(k))\} \quad k = 0, \dots, K - 1.$

Model framework



Model set

- ▶ Static, scalar map: $G \in \mathcal{R}.$
- ▶ Linear system: $y(k) = G u(k) + v(k), \quad k = 0, \dots, K - 1.$
- ▶ No disturbances: $d(k) = 0, \quad k = 0, \dots, K - 1.$
- ▶ All of the uncertainty is to be described by $v(k).$

Fitting the data

Identification procedure

Given $\mathcal{Z}$, find an estimate for G (denoted by $\hat{G}$).

We would like to pick the “best” estimate $\hat{G}$.

Fitting the experimental data

Our estimate would give a predicted output:

$$\hat{y}(k) = \hat{G} u(k), \quad k = 0, \dots, K-1.$$

Define

$$\hat{y} = \begin{bmatrix} \hat{y}(0) \\ \vdots \\ \hat{y}(K-1) \end{bmatrix}, \quad u = \begin{bmatrix} u(0) \\ \vdots \\ u(K-1) \end{bmatrix}, \quad \text{so} \quad \hat{y} = \hat{G} u.$$

$$\underset{\hat{G} \in \mathcal{R}}{\text{minimise}} \quad \|y - \hat{G}u\|_2^2$$

Fitting criteria

More than just fitting the data ...

Minimising the error in the fit addresses the fitting of the past data.

What about the predictive capabilities of the model (future data).

Would $\hat{G}$ still be a good model?

An additional assumption ...

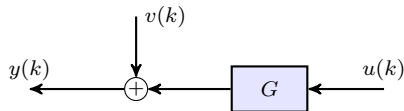
The “true plant” is an element of the model set.

This implies that there exists an optimal model: G_0

So all input-output data (future and unmeasured past) are described by G_0 .

Model class

Static, linear model with zero-mean noise



$$y(k) = G_0 u(k) + v(k), \quad \text{for all } k, \quad \text{Assumptions: } \begin{cases} G_0 \in \mathcal{R}, \\ \mathcal{E}\{v(k)\} = 0 \\ \mathcal{E}\{|v(k)|^2\} = \sigma_v^2 \end{cases}$$

Statistical statements about the model quality

- ▶ Quantify (in expectation) to ability of an estimate $\hat{G}$ to predict future outputs.
- ▶ Quantify (in expectation) the accuracy of the model with respect to the experiment length.

$\rightarrow \mathbb{E}[y - \hat{y}]$ small

$\hookrightarrow \mathbb{E}[G - \hat{G}]$ small

Identification objectives

Mean square error criterion

$$\underset{\hat{G} \in \mathcal{R}}{\text{minimise}} \mathcal{E} \left\{ \|y - \hat{G}u\|_2^2 \right\}$$

actual
output

predicted output

MSE $\rightarrow$ error

Mean
squared

error

Note that the expectation allows us to make statements about the expected square-error when a different (future) input is applied.

Identification objectives

Statistical criteria

Every identification procedure is a mapping: $\mathcal{Z} \longrightarrow \hat{G}$.

The data record $\mathcal{Z}$ is a random variable $\implies \hat{G}$ is a random variable.

Statistical criteria on $\hat{G}$:

► Bias: $\mathcal{E}\{\hat{G}\} - G$.

► Asymptotic bias:

$$\lim_{K \rightarrow \infty} \mathcal{E}\{\hat{G}\} - G.$$

► Variance: $\mathcal{E}\left\{\left|\hat{G} - \mathcal{E}\{\hat{G}\}\right|^2\right\}.$

► Consistency:

$$\lim_{K \rightarrow \infty} \hat{G}_K \xrightarrow{p} G, \text{ or equivalently, } \lim_{K \rightarrow \infty} \text{Prob}\left\{\left|\hat{G}_K - G\right| > \epsilon\right\} = 0.$$

other types of limits
as $n \rightarrow \infty$

ε70

Bias-variance trade-offs

Bias-variance relationship

Relationship: $\text{MSE} = \text{bias}^2 + \text{variance}$.

This is easily shown by multiplying out:

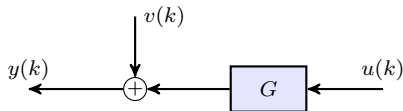
$$\begin{aligned}\text{MSE}(\hat{G}) &= \mathcal{E}\{|G - \hat{G}|^2\} \\&= \mathcal{E}\left\{\left|(G - \mathcal{E}\{\hat{G}\}) - (\hat{G} - \mathcal{E}\{\hat{G}\})\right|^2\right\} \\&= \mathcal{E}\{|G - \mathcal{E}\{\hat{G}\}|^2\} + \mathcal{E}\{|\hat{G} - \mathcal{E}\{\hat{G}\}|^2\} \\&\quad - \mathcal{E}\left\{2 \operatorname{real}\left((G - \mathcal{E}\{\hat{G}\})(\hat{G} - \mathcal{E}\{\hat{G}\})^*\right)\right\} \\&= \underbrace{\text{bias}^2(\hat{G})} + \underbrace{\text{var}(\hat{G})} \\&\quad - \underbrace{\mathcal{E}\left\{2 \operatorname{real}\left((G - \mathcal{E}\{\hat{G}\})(\hat{G} - \mathcal{E}\{\hat{G}\})^*\right)\right\}}\end{aligned}$$

The final step involves showing that the last term is zero.

Part 2: Data Averaging and Statistics

Averaging estimates

Modeling framework



R experiments:

True system: $y_r = G_0 u_r + v_r$, $v_r \sim \mathcal{N}(0, \sigma_v^2)$, v_r is i.i.d.

Each estimate comes from a single input-output pair: (y_r, u_r)

$$\hat{G}_r = \frac{y_r}{u_r} = G_0 + \frac{v_r}{u_r}, \quad r = 1, \dots, R.$$

Bias: $\mathcal{E}\{\hat{G}_r - G_0\} = 0$

Variance: $\mathcal{E}\{(\hat{G}_r - G_0)^2\} = \frac{\sigma_v^2}{|u_r|^2} = \sigma_r^2$

Averaging estimates

Averaging estimates (linear weighting)

Can we combine R unbiased estimates to get a better estimate?

$$\hat{G}_{\text{avg}} = \sum_{r=1}^R \alpha_r \hat{G}_r, \quad \alpha_r \geq 0, \quad (\text{sample mean estimate: } \alpha_r = 1/R)$$

Bias

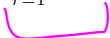
$$\text{Bias} = \mathcal{E}\{\hat{G}_{\text{avg}}\} - G_0 = \mathcal{E}\left\{\sum_{r=1}^R \alpha_r \hat{G}_r\right\} - G_0 = \left(\sum_{r=1}^R \alpha_r - 1\right) G_0.$$

The averaged estimate, $\hat{G}_{\text{avg}}$, is unbiased if and only if $\sum_{r=1}^R \alpha_r = 1$.

Why do we care about an estimate being unbiased?

Averaging estimates

Averaging estimates (linear weighting)

$$\hat{G}_{\text{avg}} = \sum_{r=1}^R \alpha_r \hat{G}_r$$


Variance

$$\begin{aligned} \text{Variance} &= \mathcal{E} \left\{ \left(\hat{G}_{\text{avg}} - \mathcal{E} \{ \hat{G}_{\text{avg}} \} \right)^2 \right\} = \mathcal{E} \left\{ \left| \sum_{r=1}^R \alpha_r \left(G + \frac{v_r}{u_r} \right) - \sum_{r=1}^R \alpha_r G \right|^2 \right\} \\ &= \mathcal{E} \left\{ \left| \sum_{r=1}^R \alpha_r \frac{v_r}{u_r} \right|^2 \right\} = \underbrace{\sum_{r=1}^R \alpha_r^2 \sigma_r^2}_{\text{pink bracket}} \\ &= \sum_{r=1}^R \alpha_r^2 \frac{\sigma_v^2}{|u_r|^2}. \end{aligned}$$

Averaging estimates

Best Linear Unbiased Estimator (BLUE)

Choose the α_r to make $\hat{G}_{\text{avg}}$ unbiased and also have the minimum variance.

Optimisation framework:

$$\begin{aligned} & \underset{\alpha_r}{\text{minimise}} && \sum_{r=1}^R \alpha_r^2 \sigma_r^2 \\ & \text{subject to} && \left(1 - \sum_{r=1}^R \alpha_r \right) = 0. \end{aligned}$$

Lagrangian: $L(\alpha_r, \lambda) = \sum_{r=1}^R \alpha_r^2 \sigma_r^2 + \lambda \left(1 - \sum_{r=1}^R \alpha_r \right).$

$$\frac{\partial L(\alpha_r, \lambda)}{\partial \alpha_r} = 2\alpha_r \sigma_r^2 - \lambda = 0, \quad \implies \quad \alpha_r = \frac{\lambda}{2\sigma_r^2}, \quad r = 1, \dots, R.$$

$$\frac{\partial L(\alpha_r, \lambda)}{\partial \lambda} = 0 \quad \implies \quad \sum_{r=1}^R \alpha_r = 1 \quad \implies \quad \lambda = 2 \left(\sum_{r=1}^R \frac{1}{\sigma_r^2} \right)^{-1}$$

Averaging estimates

Best Linear Unbiased Estimator (BLUE)

Solving for α_r gives,

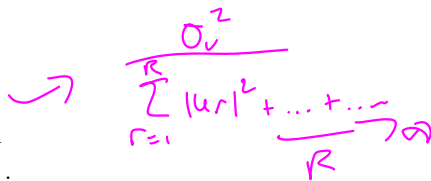
$$\alpha_r = \frac{1}{\sigma_r^2} \left(\sum_{r=1}^R \frac{1}{\sigma_r^2} \right)^{-1} = |u_r|^2 \left(\sum_{r=1}^R |u_r|^2 \right)^{-1}.$$

The minimum variance weights are proportional to the inverse of the variance.

Low variance estimates have greater weight.

The resulting variance is:

$$\text{var}(\hat{G}_{\text{avg}}) = \left(\sum_{r=1}^R \sigma_r^{-2} \right)^{-1} = \underline{\sigma_v^2} \left(\sum_{r=1}^R |u_r|^2 \right)^{-1}.$$

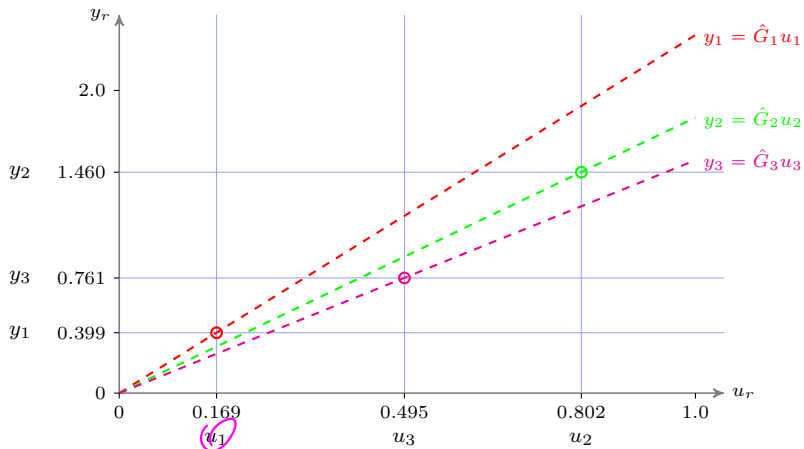


Handwritten diagram illustrating the variance formula. It shows the expression $\frac{\sigma_v^2}{\sum_{r=1}^R |u_r|^2 + \dots + \dots}$ with a pink arrow pointing from the text 'The resulting variance is:' to it. The denominator is further annotated with a pink line and the letter 'R' below it, and a pink arrow pointing to the right, indicating that the denominator increases as R increases.

The variance decreases as $R \rightarrow \infty$, implying that the estimator is **consistent**.

Example

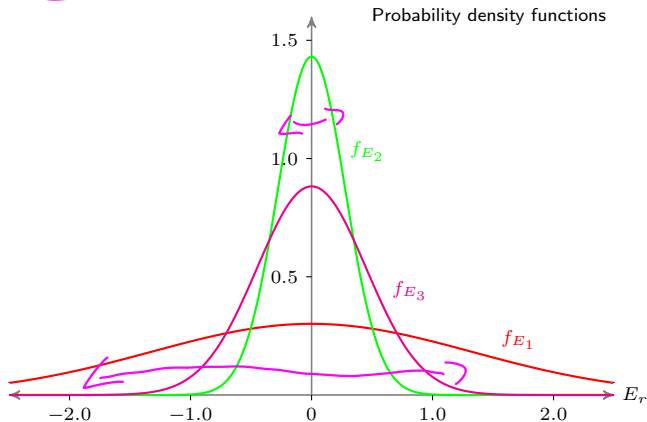
$$y = G_0 u + v, \quad v \sim \mathcal{N}(0, 0.05), \quad (\sigma_v^2 = 0.05)$$



Example

Error distributions: $E_r = \hat{G}_r - G_0$

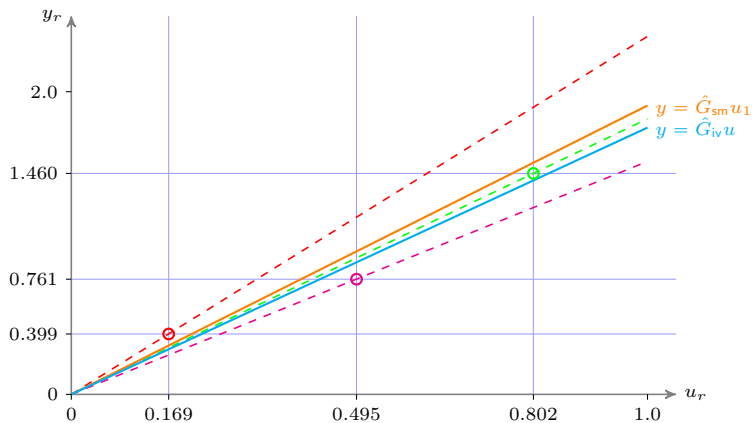
$$\mathcal{E}\{E_r\} = 0, \quad \sigma_{E_r}^2 = \frac{\sigma_v^2}{|u_r|^2}, \quad r = 1, 2, 3.$$



Example – Averaging: estimated models

Sample mean $\hat{G}_{\text{sm}} : \alpha_r = \frac{1}{R}$, Inverse variance $\hat{G}_{\text{iv}} : \alpha_r = \frac{1/\sigma_r^2}{\sum_{r=1}^R 1/\sigma_r^2}$

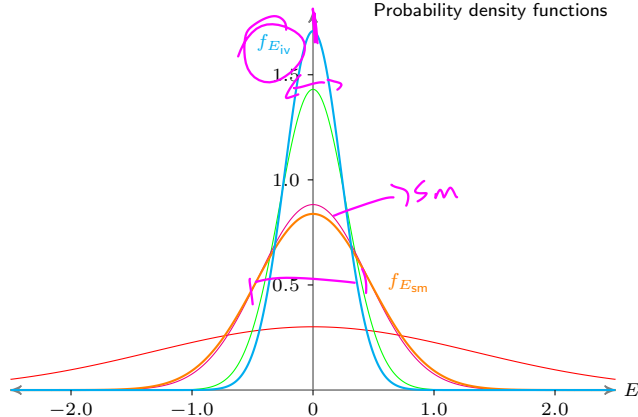
$\hat{G}_{\text{sm}} = 1.9087$ $\hat{G}_{\text{iv}} = 1.7627$



Example – Averaging: error variances

Sample mean $\hat{G}_{\text{sm}} : \alpha_r = \frac{1}{R}$, Inverse variance $\hat{G}_{\text{iv}} : \alpha_r = \frac{1/\sigma_r^2}{\sum_{r=1}^R 1/\sigma_r^2}$.

Probability density functions



$$\sigma_{\text{sm}}^2 = 0.2266$$

$$\sigma_{\text{iv}}^2 = 0.0546$$

Averaging estimates

Minimising the Mean-Square Error (MSE)

Does unbiased and minimum variance imply minimum MSE?

$$\text{MSE}(\hat{G}_{\text{avg}}) = \text{bias}^2 + \text{variance} = \left(1 - \sum_{r=1}^R \alpha_r\right)^2 |G|^2 + \sum_{r=1}^R \alpha_r^2 \sigma_r^2.$$

Handwritten note: $\rightarrow \rho \| \hat{G} \|^2$

Minimising this over $\alpha_r > 0$ gives,

$$\alpha_r = \frac{\frac{1}{\sigma_r^2}}{\frac{1}{|G|^2} + \sum_{r=1}^R \frac{1}{\sigma_r^2}} = \frac{|u_r|^2}{\frac{\sigma_v^2}{|G|^2} + \sum_{r=1}^R |u_r|^2}.$$

The optimal weighting, α_r , for minimising the MSE depends on σ_v^2 and $|G|^2$.

Averaging estimates

Minimising the Mean-Square Error (MSE)

Using the optimal α_r weighting gives the best possible MSE,

$$\text{MSE}(\hat{G}_{\text{avg}}) = : \frac{1}{\frac{1}{|G|^2} + \sum_{r=1}^R \frac{1}{\sigma_r^2}} = \frac{\sigma_v^2}{\frac{\sigma_v^2}{|G|^2} + \sum_{r=1}^R |u_r|^2}.$$

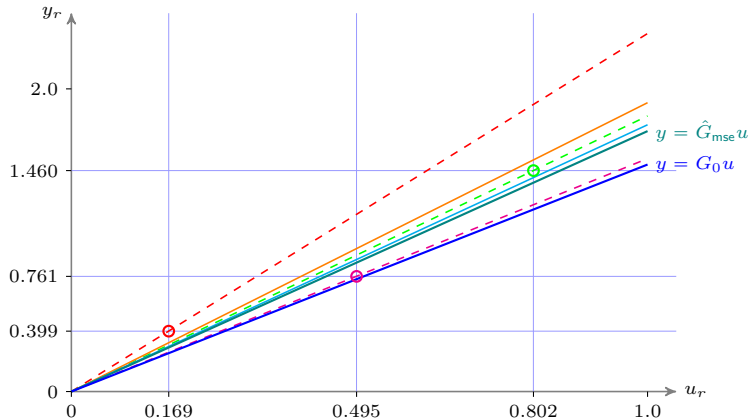
as $R \rightarrow \infty$, effect of G term $\rightarrow 0$

This is strictly less than the MSE for the minimum variance estimate.

Example – Averaging: error variances

Minimum MSE $\hat{G}_{\text{mse}} : \alpha_r = \frac{1/\sigma_r^2}{\sigma_v^2/|G_0|^2 + \sum_{r=1}^R 1/\sigma_r^2} \cdot$

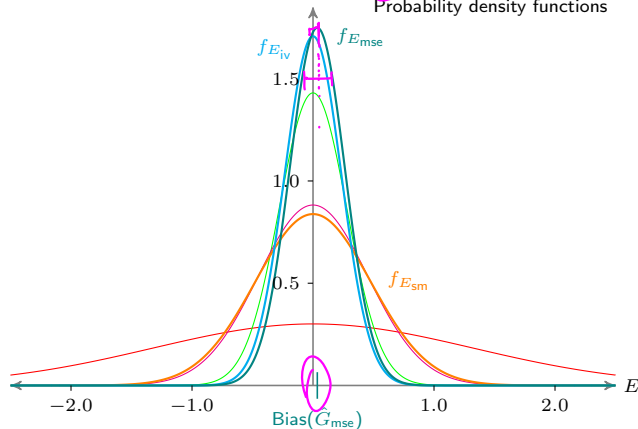
$\hat{G}_{\text{sm}} = 1.9087,$ $\hat{G}_{\text{iv}} = 1.7627,$ $\hat{G}_{\text{mse}} = 1.7209,$ $G_0 = 1.5$



Example – Averaging: error variances

Minimum MSE $\hat{G}_{\text{mse}}$: $\alpha_r = \frac{1/\sigma_r^2}{\sigma_v^2/|\hat{G}_0|^2 + \sum_{r=1}^R 1/\sigma_r^2}$.

Probability density functions



$$\sigma_{\text{sm}}^2 = 0.2266$$

$$\sigma_{\text{iv}}^2 = 0.0546$$

$$\sigma_{\text{mse}}^2 = 0.0520$$

$$\text{Bias}(\hat{G}_{\text{mse}}) = 0.0355$$

$$\text{MSE}(\hat{G}_{\text{mse}}) = 0.0533$$

Averaging estimates

Bias-variance trade-off

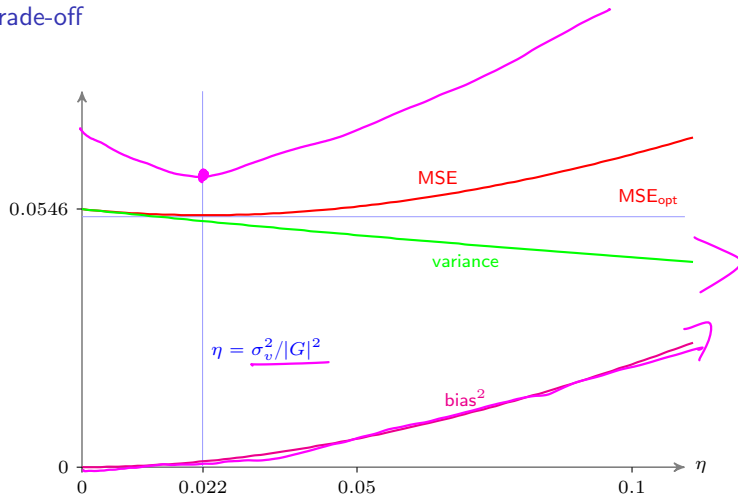
Parametrised family of estimates:

$$\alpha_r = \frac{|u_r|^2}{\eta + \sum_{r=1}^R |u_r|^2}, \quad \eta \geq 0. \quad (\eta \text{ is a hyperparameter})$$

Ideally $\eta = \sigma_v^2/|G|^2$ and $\eta = 0$ corresponds to the minimum variance (unbiased) estimate.

Example

Bias-Variance trade-off



Averaging estimates

Picking η : validation

Averaging estimates: summary

$$\hat{G}_{\text{avg}} = \sum_{r=1}^R \alpha_r \hat{G}_r, \quad \alpha_r \geq 0,$$

Bias and Variance of $\hat{G}_{\text{avg}}$:

$$\text{Bias} = \left(\sum_{r=1}^R \alpha_r - 1 \right) G_0,$$

$$\text{Variance} = \sum_{r=1}^R \alpha_r^2 \sigma_r^2.$$

Weights: α_r

Inverse variance weighting	Minimum MSE weighting
$\alpha_r = \frac{1}{\sigma_r^2} \left(\sum_{r=1}^R \frac{1}{\sigma_r^2} \right)^{-1}$	$\alpha_r = \frac{1}{\sigma_r^2} \left(\frac{1}{ G ^2} + \sum_{r=1}^R \frac{1}{\sigma_r^2} \right)^{-1}$
unbiased	biased (finite R)
Best Linear Unbiased Estimator	asymptotically unbiased
variance decrease order $1/R$	minimum MSE
	variance decrease order $1/R$

Parameter estimation statistics

Basic formulation

Model parameters: $\theta \in \mathcal{M}^d$

True parameter: $\theta_0 \rightarrow$ in the model set

Estimated parameter: $\hat{\theta}$

Maximum likelihood estimation

Basic formulation

Consider K observations, $z_1, \dots, z_K$.

Each is a realisation of a random variable, with joint probability distribution,

$f(\underbrace{x_1, \dots, x_K}_{\text{random variables}}; \theta)$ $\leftarrow$ family of distributions parametrised by θ .

$N(\mu, \sigma)$

Another common notation is,

$f(x_1, \dots, x_K | \theta)$ $\leftarrow$ the pdf for $x_1, \dots, x_K$ given θ .

For independent variables,

$$\underbrace{f(x_1, \dots, x_K; \theta)} = \underbrace{f_1(x_1; \theta)} \underbrace{f_2(x_2; \theta)} \cdots \underbrace{f_K(x_K; \theta)} = \prod_{i=1}^K f_i(x_i; \theta)$$

Maximum likelihood estimation

Likelihood function

Substituting the observation, $Z_K = \{z_1, \dots, z_K\}$, gives a function of θ ,

$$\mathcal{L}(\theta) = f(x_1, \dots, x_K; \theta) \Big|_{x_i = z_i, i=1, \dots, K.} \quad (\text{Likelihood function})$$

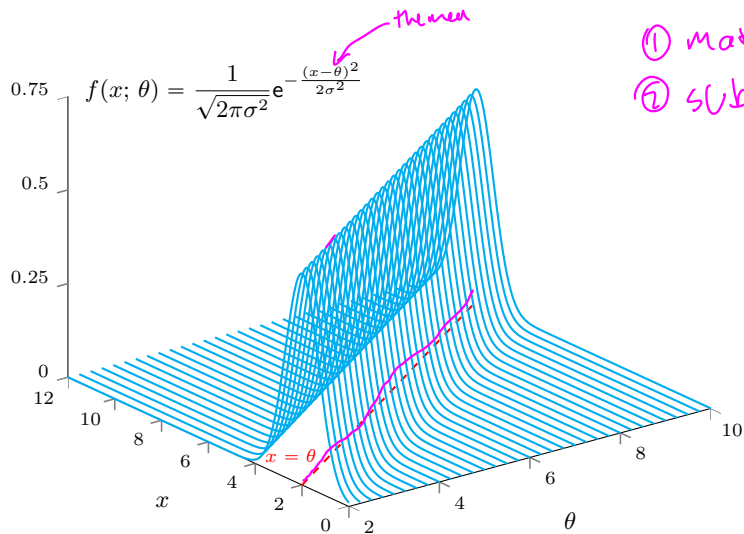
Maximum likelihood estimator:

$$\hat{\theta}_{\text{ML}} = \underset{\theta}{\operatorname{argmax}} \mathcal{L}(\theta).$$

The value chosen for θ is the one that gives the most “agreement” with the observation.

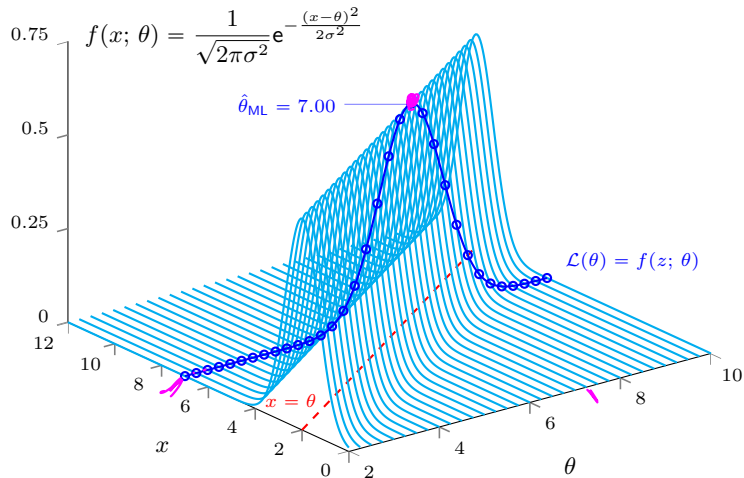
Maximum likelihood estimation

Estimating the mean of a Gaussian distribution ($\sigma^2 = 0.5$)



Maximum likelihood estimation

Estimating the mean of a Gaussian distribution ($\sigma^2 = 0.5$) Datum: $z = 7.0$



Maximum likelihood estimation

Log-likelihood function

It is often mathematically easier to consider,

$$\hat{\theta}_{\text{ML}} = \underset{\theta}{\operatorname{argmax}} \underbrace{\ln \mathcal{L}(\theta)}.$$

As the $\ln$ function is monotonic this gives the same θ .

This is typically the natural logarithm so as to be able to handle the exponentiation in typical pdfs.

Example

Estimation of the mean of a set of samples

$$z_i, \quad i = 1, \dots, K \quad z_i \sim \mathcal{N}(\theta_0, \sigma_i^2). \quad (\text{note: different variances})$$

Sample mean estimate: $\hat{\theta}_{\text{SM}} = \frac{1}{K} \sum_{i=1}^K z_i$

Probability density functions (pdf): θ is the common mean of the distributions.

$$f_i(x_i; \theta) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x_i - \theta)^2}{2\sigma_i^2}\right)$$

For independent samples the joint pdf is:

$$f(x_1, \dots, x_K; \theta) = \prod_{i=1}^K \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x_i - \theta)^2}{2\sigma_i^2}\right)$$

Example

Estimation of the mean of a set of samples

$$\theta_{\text{ML}} = \operatorname{argmax}_{\theta} \ln f(x_1, \dots, x_K; \theta) \Big|_{x_i = z_i, i=1, \dots, K}.$$

$$= \operatorname{argmax}_{\theta} \ln \mathcal{L}(\theta)$$

$$= \operatorname{argmax}_{\theta} \left(\underbrace{-\frac{K}{2} \ln(2\pi)}_{\text{constant}} - \underbrace{\sum_{i=1}^K \frac{1}{2} \ln(\sigma_i^2)}_{\text{constant}} - \underbrace{\frac{1}{2} \sum_{i=1}^K \frac{(z_i - \theta)^2}{\sigma_i^2}}_{\text{depends on } \theta} \right)$$

This gives (differentiate and equate to zero),

$$\hat{\theta}_{\text{ML}} = \left(\frac{1}{\sum_{i=1}^K \frac{1}{\sigma_i^2}} \right) \sum_{i=1}^K \frac{z_i}{\sigma_i^2}$$

go to zero when you take derivatives

Bayesian approach

Random parameter framework

Consider θ to be a random variable with pdf: $f_{\theta}(x)$.

This is an **a priori** distribution (assumed before the experiment).



Thomas Bayes
1701–1761.

Conditional distribution (inference from the experiment)

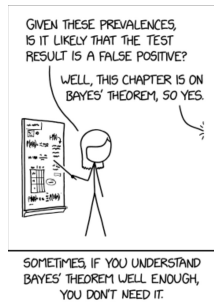
Our model (plus assumptions) gives a conditional distribution,

$$f(x_1, \dots, x_K \mid \theta)$$

On the basis of the experiment ($x_i = z_i$),

$$\text{Prob}(\theta \mid z_1, \dots, z_K) = \frac{\text{Prob}(Z_K \mid \theta) \text{Prob}(\theta)}{\text{Prob}(Z_K)}$$

So, $\text{argmax}_{\theta} f(\theta \mid z_1, \dots, z_K) = \text{argmax}_{\theta} f(Z_K \mid \theta) f_{\theta}(\theta)$



©XKCD, CC-BY-NC 2.5

Maximum a posteriori (MAP) estimation

Estimator

Given data, Z_K ,

$$\hat{\theta}_{\text{MAP}} = \underset{\theta}{\operatorname{argmax}} f(Z_K | \theta) f_{\theta}(\theta).$$

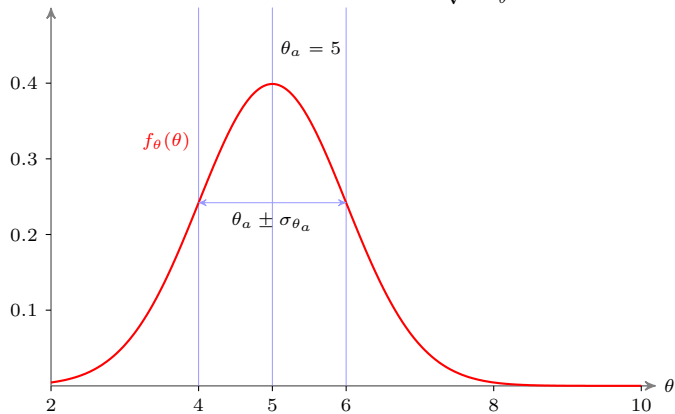
We can interpret the maximum likelihood estimator as,

$$\begin{aligned} \theta_{\text{ML}} &= \underset{\theta}{\operatorname{argmax}} f(x_1, \dots, x_K; \theta) \Big|_{x_i = z_i, i=1, \dots, K}. \\ &= \underset{\theta}{\operatorname{argmax}} f(Z_K | \theta) \end{aligned}$$

These estimates coincide if we assume a uniform distribution for θ .

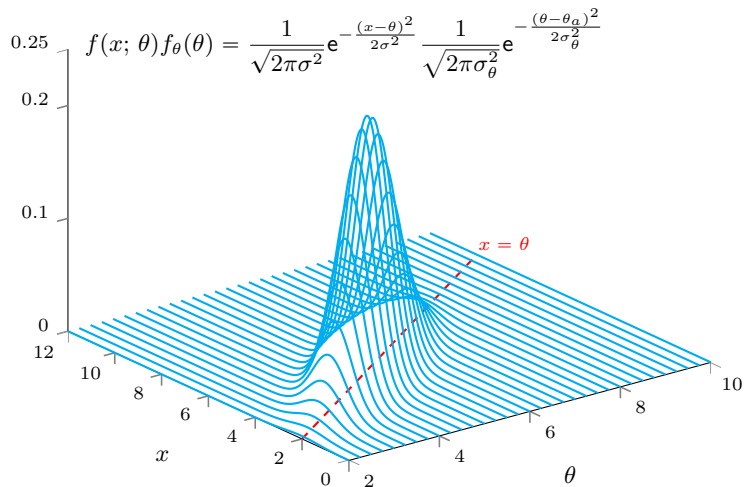
MAP estimation

A priori parameter distribution: $f_{\theta}(\theta) = \frac{1}{\sqrt{2\pi\sigma_{\theta}^2}} e^{-\frac{(\theta-\theta_a)^2}{2\sigma_{\theta}^2}}$, $\theta_a = 5$, $\sigma_{\theta}^2 = 1$.



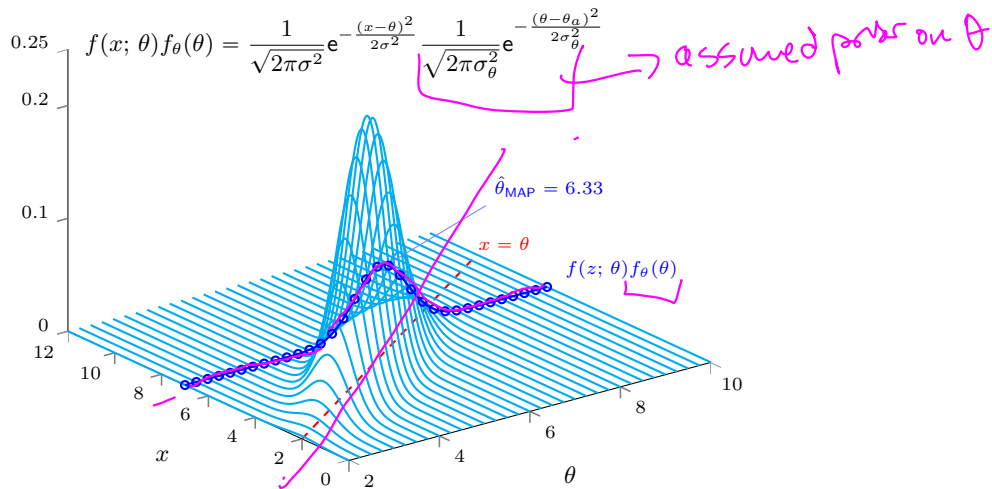
MAP estimation

Estimating the mean: Gaussian distribution ($\sigma^2 = 0.5$, $\theta_a = 5$, $\sigma_{\theta_a}^2 = 1$)



MAP estimation

Estimating the mean: Gaussian distribution ($\sigma^2 = 0.5$, $\theta_a = 5$, $\sigma_{\theta_a}^2 = 1$) Datum: $z = 7.0$



Cramér-Rao bound

Mean-square error matrix

$$P = \mathcal{E} \left\{ \left(\hat{\theta}(\mathcal{Z}_K) - \theta_0 \right) \left(\hat{\theta}(\mathcal{Z}_K) - \theta_0 \right)^T \right\}$$

Assume that the pdf for $\mathcal{Z}_K$ is $f(\mathcal{Z}_K; \theta)$.

Cramér-Rao inequality

Assume $\mathcal{E} \left\{ \hat{\theta}(\mathcal{Z}_K) \right\} = \theta_0$, and $\mathcal{Z}_K \subset \mathcal{R}^K$.

Then, $P \geqslant M^{-1}$ (M is the Fisher Information Matrix)

$$\begin{aligned} M &= \mathcal{E} \left\{ \left(\frac{d}{d\theta} \ln f(\mathcal{Z}_K; \theta) \right) \left(\frac{d}{d\theta} \ln f(\mathcal{Z}_K; \theta) \right)^T \right\} \Big|_{\theta=\theta_0} \\ &= - \mathcal{E} \left\{ \frac{d^2}{d\theta^2} \ln f(\mathcal{Z}_K; \theta) \right\} \Big|_{\theta=\theta_0} \end{aligned}$$

Maximum likelihood: statistical properties

Asymptotic results for i.i.d. variables

Consider a parametrised family of pdfs,

$$f(x_1, \dots, x_K; \theta) = \prod_{i=1}^K f_i(x_i; \theta).$$

Then,

$$\lim_{K \rightarrow \infty} \hat{\theta}_{\text{ML}} \xrightarrow{\text{w.p. 1}} \theta_0,$$

and

$$\lim_{K \rightarrow \infty} \sqrt{K} \left(\hat{\theta}_{\text{ML}}(\mathcal{Z}_K) - \theta_0 \right) \sim \mathcal{N}(0, M^{-1}).$$

another type of limit

Next Time

- 
- ▶ Pulse response
 - ▶ Model-predictive control
 - ▶ Data-enabled predictive control (DeePC)



